

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA1012	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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Date:	5/08/2026	Duration:	60 minutes

Code of Conduct

I T. Niranjan Kumar (192473007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Niranjan

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, **analyze** the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. **Analyze** System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze** the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and	Diagram is incomplete, unclear, or

1.1 Problem Statement

The open-source software ecosystem has become one of the most important platforms for software development and innovation. Millions of developers contribute to open-source projects hosted on platforms such as GitHub to improve software quality, gain practical experience, and collaborate with global developer communities. For beginner developers, contributing to open-source projects is an excellent opportunity to develop technical skills, build a professional portfolio, and gain industry exposure. However, identifying suitable projects remains a major challenge due to the vast number of repositories available, varying project complexities, and the lack of personalized guidance.

Existing repository search mechanisms are primarily keyword-based and do not consider a developer's programming knowledge, interests, or experience level. As a result, beginners often spend considerable time searching for repositories that match their abilities. They also face difficulties in tracking contributions across multiple repositories, monitoring pull request status, understanding contribution guidelines, and measuring their overall learning progress.

To address these challenges, the proposed **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** provides intelligent repository recommendations, contribution tracking, project difficulty classification, portfolio generation, and personalized notifications. The system simplifies the contribution process and encourages continuous participation in open-source communities by connecting beginners with suitable projects based on their skills and interests.

1.2 System Objectives

The proposed system aims to achieve the following objectives:

- Recommend beginner-friendly open-source repositories based on the user's programming languages, interests, and experience level.
- Integrate with GitHub API to retrieve repositories containing beginner-friendly labels such as **Good First Issue** and **Help Wanted**.
- Track pull requests, commits, issues, and contribution history from multiple repositories through a centralized dashboard.
- Measure developer skill progression using contribution frequency, repository diversity, and technology exposure.
- Classify repositories according to their estimated difficulty to assist beginners in selecting suitable projects.
- Notify users whenever new repositories or issues matching their technical profile become available.
- Generate a professional contribution portfolio suitable for resumes and job applications.

- Encourage continuous learning and active participation within open-source communities.

1.3 Functional Requirements

Functional requirements describe the core services and operations that the proposed system must perform to satisfy user needs. These requirements define the expected behavior of the application and ensure that all essential features are successfully implemented.

Table 1. Functional Requirements

Requirement ID	Functional Requirement	Description
FR1	User Authentication	Allows users to register, log in, and securely access the platform.
FR2	Skill Profile Management	Enables users to specify programming languages, interests, and experience level.
FR3	GitHub Integration	Connects with GitHub API to retrieve repositories, issues, and contribution data.
FR4	Repository Recommendation	Suggests beginner-friendly repositories based on user skills and preferences.
FR5	Contribution Tracking	Tracks pull requests, commits, issue participation, and merge status.
FR6	Progress Dashboard	Displays contribution statistics, technology diversity, and skill progression.
FR7	Portfolio Generation	Creates a professional summary of open-source contributions.
FR8	Notification System	Sends alerts about new beginner-friendly issues matching the user's profile.

1.4 Non-Functional Requirements

Non-functional requirements define the quality standards that the software must satisfy. These requirements ensure that the application remains efficient, secure, reliable, and scalable while providing a positive user experience.

Table 2. Non-Functional Requirements

Requirement ID	Non-Functional Requirement	Description
NFR1	Performance	The system should provide repository recommendations with minimal response time.
NFR2	Scalability	The application should support a growing number of users and repositories efficiently.
NFR3	Security	User credentials and GitHub authentication tokens must be securely protected.
NFR4	Reliability	The platform should provide continuous and accurate services with minimal failures.
NFR5	Maintainability	The modular architecture should simplify updates, bug fixes, and feature additions.
NFR6	Availability	The application should remain accessible with high uptime and minimal service interruptions.
NFR7	Usability	The user interface should be intuitive, responsive, and beginner-friendly.

1.5 Quality Attributes

Scalability

The proposed system is designed to support an increasing number of users, repositories, and GitHub API requests without affecting overall performance. Its modular architecture allows additional features and services to be integrated easily as the platform grows.

Security

Security is a critical aspect of the system. GitHub authentication is implemented using secure OAuth mechanisms, while user information and access tokens are protected through encryption and secure communication protocols to prevent unauthorized access.

Performance

The recommendation engine and contribution tracker are optimized to deliver repository suggestions, dashboard updates, and contribution statistics with minimal response time, ensuring a smooth user experience.

Reliability

The system is expected to provide consistent recommendations and accurate contribution tracking while maintaining stable operation with minimal downtime. Proper error handling and data validation techniques improve the reliability of the platform.

Maintainability

The application follows a modular software architecture that separates different functional components. This design simplifies debugging, testing, maintenance, and future feature enhancements without affecting other modules.

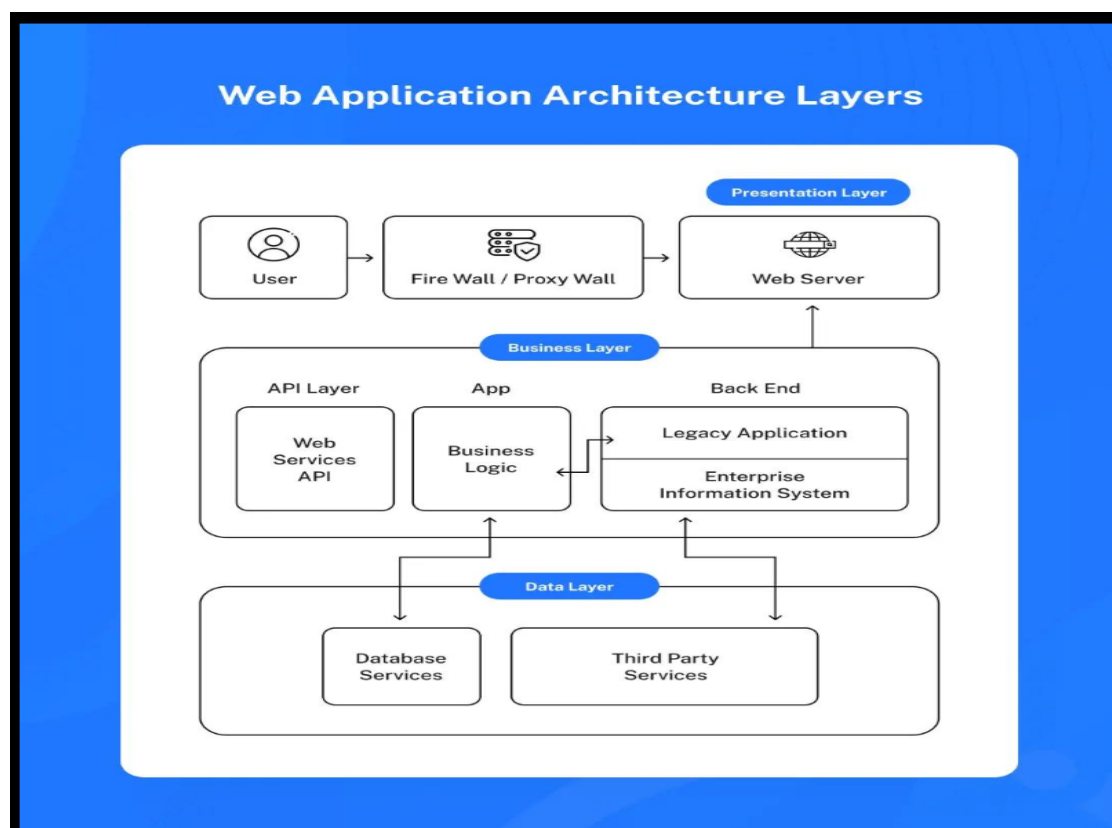
Availability

The platform is designed to remain operational throughout the day, enabling users to access repository recommendations, dashboards, and contribution tracking services whenever required. High availability improves user satisfaction and encourages continuous participation in open-source development.

2. Select a Suitable Software Architecture

Selecting an appropriate software architecture is an important step in software system development because it determines how different components interact, how the system handles user requests, and how easily it can be maintained or expanded in the future. The proposed **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** requires an architecture that supports modular development, external API integration, secure communication, efficient data management, and future scalability.

After analyzing the system requirements, **Layered Architecture** has been selected as the most suitable architectural style for this project.



2.1 Selected Software Architecture

The proposed system follows a **Layered Architecture**, where the application is divided into multiple independent layers. Each layer has a specific responsibility and communicates only with its adjacent layers. This separation of responsibilities improves code organization, simplifies maintenance, and allows individual components to be modified without affecting the entire system.

The major layers of the proposed architecture are:

- **Presentation Layer (Frontend):** Provides the graphical user interface through which users register, log in, view recommendations, track contributions, and generate portfolios.
- **Application Layer (Backend):** Processes user requests, implements business logic, validates data, and coordinates communication between different modules.
- **Business Logic Layer:** Contains the recommendation engine, contribution tracker, project difficulty scoring module, portfolio generator, and notification manager.
- **Data Access Layer:** Manages communication with the database and GitHub API, retrieves repository information, stores user data, and updates contribution records.
- **Database Layer:** Stores user profiles, contribution history, recommendation results, notification records, and application settings.

2.2 Why Layered Architecture?

Layered Architecture is selected because it provides a simple, organized, and scalable design suitable for web-based applications. Since the proposed system consists of multiple independent modules such as user management, recommendation engine, GitHub integration, contribution tracking, and portfolio generation, separating these functionalities into different layers improves maintainability and reduces system complexity.

Another important advantage is that each layer performs a dedicated responsibility. The user interface is separated from business logic, while database operations are handled independently. This modular approach allows developers to update or replace one layer without affecting the remaining layers.

Layered Architecture also simplifies testing because each layer can be verified independently. Future enhancements such as AI-based recommendations, machine learning models, or additional repository platforms can be integrated without redesigning the complete application.

2.3 Justification for Selecting Layered Architecture

The selected architecture satisfies both the functional and non-functional requirements identified during system analysis. It supports secure GitHub API communication, centralized contribution tracking, personalized repository recommendations, and efficient database management.

The architecture also provides excellent maintainability because developers can independently modify the frontend, backend, recommendation engine, or database without introducing major changes to the remaining modules. This modular structure reduces development effort and simplifies debugging.

Furthermore, Layered Architecture provides better scalability. As the number of users and repositories increases, individual layers can be optimized or upgraded independently to handle additional workload. The architecture also improves code readability, reusability, and long-term software maintenance, making it highly suitable for educational and industrial applications.

2.4 Comparison with Other Software Architectures

Table 3. Comparison of Software Architectures

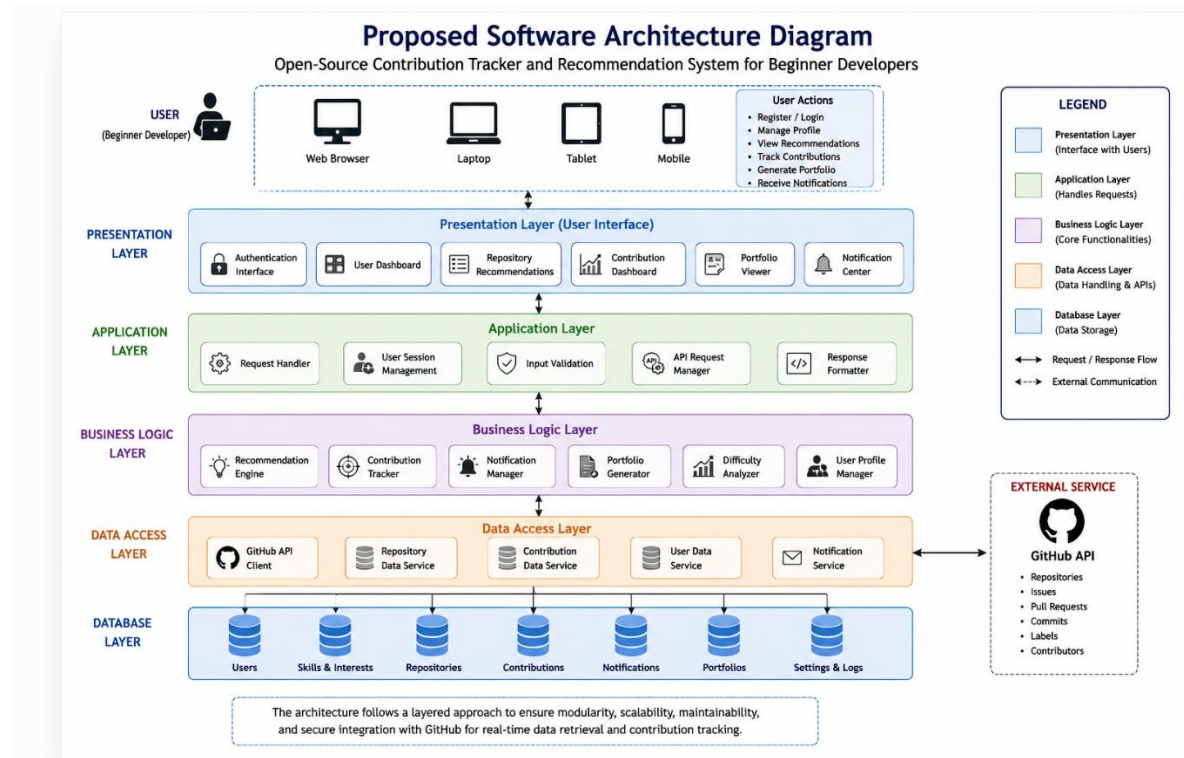
Architecture Style	Advantages	Limitations	Suitability for this Project
Layered Architecture	Modular design, easy maintenance, secure, scalable, simple testing	Slight communication overhead between layers	Highly Suitable (Selected)
MVC Architecture	Good separation between UI and business logic	Mainly focuses on presentation design rather than complete system organization	Suitable but less flexible than Layered Architecture
Microservices Architecture	Highly scalable and independently deployable services	Complex deployment, higher infrastructure cost, difficult for beginner-level projects	Not suitable for this project due to increased complexity

2.5 Summary

Based on the system requirements, **Layered Architecture** is selected as the most appropriate architectural style for the proposed system. It provides a well-structured, modular, secure, and maintainable framework capable of integrating GitHub APIs, recommendation modules, contribution tracking services, and database operations efficiently. The architecture also

supports future enhancements while maintaining high performance, scalability, and reliability, making it an ideal choice for the **Open-Source Contribution Tracker and Recommendation System for Beginner Developers**.

1. Draw the Software Architecture Diagram



3.1 Software Architecture Diagram

The proposed **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** follows a **Layered Architecture** that separates the application into different functional layers. Each layer has a specific responsibility and communicates only with adjacent layers. This modular design improves maintainability, scalability, security, and overall system performance.

The architecture consists of five major layers: the **Presentation Layer**, **Application Layer**, **Business Logic Layer**, **Data Access Layer**, and **Database Layer**. The Presentation Layer provides the user interface through which developers interact with the system. The Application Layer processes user requests and coordinates communication between different modules. The Business Logic Layer performs recommendation generation, contribution tracking, portfolio management, difficulty scoring, and notification handling. The Data Access Layer manages communication with GitHub APIs and the database, while the Database Layer stores user profiles, repository information, contribution history, notifications, and application settings.

3.2 Architecture Description

The architecture begins with the **Beginner Developer**, who accesses the application using a web browser. The user logs into the system and provides programming skills, interests, and GitHub account details. These requests are handled by the **Presentation Layer**, which forwards them to the **Application Layer**.

The Application Layer validates the user's requests and forwards them to the appropriate business modules. The **Recommendation Engine** retrieves beginner-friendly repositories using the GitHub API and filters them according to the user's programming languages and interests. The **Contribution Tracker** monitors pull requests, issues, commits, and merge status from multiple repositories. The **Portfolio Generator** summarizes the user's achievements into a professional portfolio, while the **Notification Manager** continuously checks for new beginner-friendly issues and sends alerts whenever matching opportunities become available.

The Data Access Layer communicates with the GitHub API to retrieve repository information and stores relevant data in the database. The Database Layer maintains user profiles, recommendation history, contribution records, notification logs, and application configurations. Finally, the processed information is sent back to the Presentation Layer, where it is displayed through dashboards, progress reports, recommendations, and portfolio summaries.

3.3 Advantages of the Proposed Architecture

The selected architecture provides several benefits for the proposed system:

- It separates user interface, business logic, and database operations into independent layers.
- It improves maintainability by allowing each module to be updated independently.
- It supports scalability by enabling the system to handle an increasing number of users and repositories.
- It enhances security by isolating authentication and API communication from the presentation layer.
- It simplifies debugging, testing, and future feature integration.
- It efficiently integrates with GitHub APIs while maintaining a modular software structure.

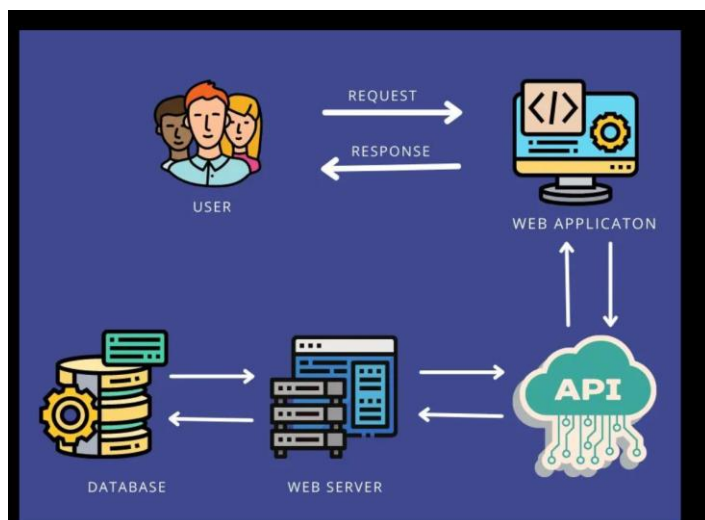
4.1 Component Interactions

- The interaction between the components of the proposed system is as follows:
- The user logs into the application using GitHub authentication.
- The Presentation Layer sends the request to the Application Layer.

- The Application Layer validates the user information.
- The Recommendation Engine retrieves beginner-friendly repositories through the GitHub API.
- The Contribution Tracker monitors pull requests, commits, and issue activities.
- The Database stores user profiles, contribution history, and recommendation data.
- The Notification System informs users about newly available beginner-friendly issues.
- The updated information is displayed on the user dashboard.

4.2 Overall Workflow

- The workflow of the proposed system is shown below:
- User Registration/Login.
- Skill Profile Creation.
- GitHub Authentication.
- Repository Retrieval using GitHub API.
- Repository Recommendation.
- User Contribution.
- Contribution Tracking.
- Progress Dashboard Update.
- Portfolio Generation.
- Notification of New Issues



5. Discuss Quality Attributes

Software quality attributes define the characteristics that determine how effectively the proposed system performs under different operating conditions. The selected Layered Architecture supports several important quality attributes that improve the overall performance, reliability, and usability of the application.

5.1 Scalability

Scalability refers to the ability of the system to handle an increasing number of users, repositories, and GitHub API requests without reducing system performance. The proposed architecture separates each functional module into independent layers, allowing additional servers, databases, or services to be integrated whenever system demand increases. This enables the application to support future growth efficiently.

5.2 Security

Security is one of the most critical quality attributes of the proposed system. User authentication is implemented using GitHub OAuth authentication, ensuring secure login without exposing user credentials. Communication between the application and GitHub API is protected using HTTPS protocols. User information, authentication tokens, and contribution history are securely stored within the database, minimizing the risk of unauthorized access.

5.3 Performance

The proposed system is designed to provide fast repository recommendations and real-time contribution tracking. Efficient database queries, optimized API communication, and modular processing help reduce response time while ensuring that users receive accurate recommendations and dashboard updates with minimal delay.

5.4 Reliability

Reliability ensures that the system continuously provides accurate recommendations and contribution tracking services without unexpected failures. Proper exception handling, input validation, and secure API communication improve system stability. Even if external API requests fail temporarily, stored data remains available, ensuring uninterrupted user experience.

5.5 Maintainability

Maintainability refers to the ease with which developers can modify, update, and extend the application. Since the proposed system follows a Layered Architecture, each module performs

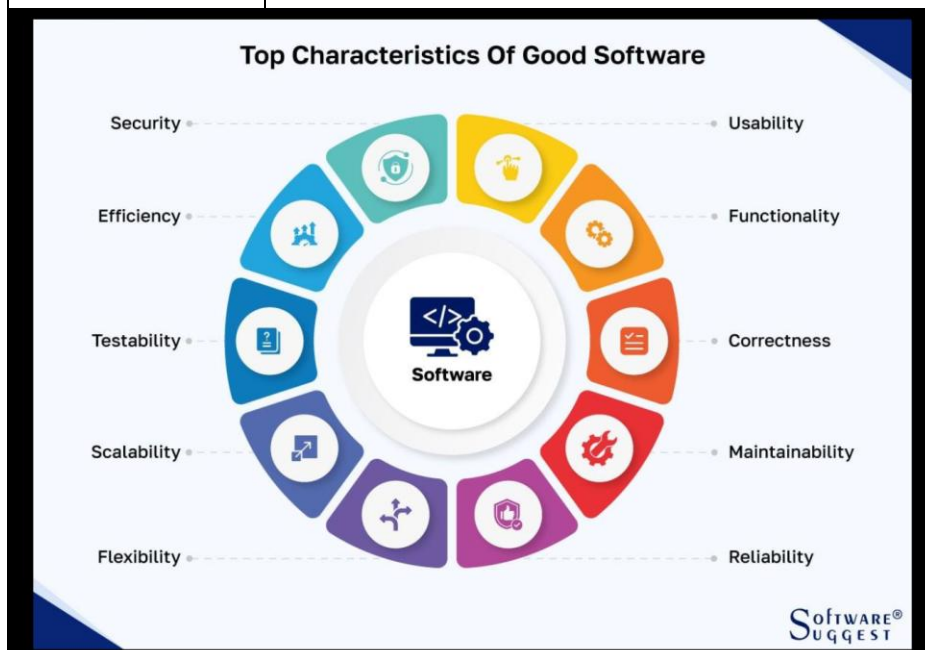
a specific responsibility independently. This modular design simplifies debugging, testing, and feature enhancements without affecting the remaining components.

5.6 Availability

Availability ensures that the proposed platform remains operational whenever users need to access repository recommendations, contribution history, or portfolio generation services. The application is designed to minimize downtime through proper error handling, reliable database management, and continuous monitoring of system resources.

Table 5. Quality Attribute Evaluation

Quality Attribute	Implementation in Proposed System
Scalability	Supports increasing users and repositories through modular architecture.
Security	Uses GitHub OAuth authentication, HTTPS communication, and secure data storage.
Performance	Optimized API communication and efficient database operations reduce response time.
Reliability	Provides accurate recommendations with continuous system availability.
Maintainability	Independent layers simplify updates, testing, and debugging.
Availability	Ensures users can access the application with minimal downtime.



6. Justify Architectural Decisions

6.1 Reasons for Selecting Layered Architecture

The proposed **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** adopts a **Layered Architecture** because it provides a clear separation between user interface, application processing, business logic, and database management. This separation improves code organization, simplifies software development, and enhances long-term maintainability.

Layered Architecture also allows developers to modify or replace individual components without affecting the remaining parts of the application. This flexibility makes the architecture suitable for future feature additions such as artificial intelligence-based recommendations, advanced analytics, and integration with additional code hosting platforms.

6.2 Architectural Trade-offs

Although Layered Architecture offers several advantages, it also involves certain trade-offs. Communication between layers introduces a small processing overhead compared to tightly coupled systems. However, this overhead is minimal when compared with the significant benefits gained in maintainability, scalability, security, and modularity.

For the proposed system, these advantages greatly outweigh the minor performance overhead, making Layered Architecture the most suitable architectural choice.

6.3 Future Enhancements

The proposed architecture provides an excellent foundation for future improvements. Additional features that can be integrated include:

- Artificial Intelligence-based repository recommendation.
- Machine Learning algorithms for personalized skill prediction.
- Support for GitLab and Bitbucket repositories.
- Real-time collaboration analytics.
- Mobile application support.
- Advanced portfolio analytics.
- Community reputation scoring.
- AI-powered mentor recommendation system.

Because of the modular architecture, these enhancements can be implemented without major modifications to the existing system.

6.4 Long-Term Maintainability

The modular structure of Layered Architecture improves long-term software maintenance by allowing independent updates to individual components. Developers can modify the recommendation engine, user interface, notification system, or database without affecting the remaining modules. This significantly reduces maintenance cost, improves software quality, and extends the operational life of the application.

6.5 System Integration

The proposed architecture supports seamless integration with external services through well-defined interfaces. GitHub API integration enables real-time repository retrieval, contribution tracking, and issue monitoring. Additional APIs and cloud services can also be integrated in the future without redesigning the overall architecture, making the system highly flexible and extensible.

Conclusion

The **Open-Source Contribution Tracker and Recommendation System for Beginner Developers** provides an effective solution for simplifying the open-source contribution process. By combining personalized repository recommendations, GitHub integration, contribution tracking, notification services, and portfolio generation within a Layered Architecture, the proposed system successfully addresses the challenges faced by beginner developers. The architecture ensures scalability, security, reliability, maintainability, and high performance while supporting future enhancements and long-term software evolution. Overall, the proposed system serves as a comprehensive platform that encourages continuous learning, improves developer engagement, and strengthens participation in the global open-source community.

References

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